

C-Band Silicon Photonics Mach–Zehnder and Michelson Interferometers: Design, Fabrication, and Experimental Analysis

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Abstract— We report the design, fabrication, and characterization of compact C-band Mach–Zehnder and Michelson interferometers on a silicon-on-insulator (SOI) photonic integrated circuit (PIC) platform. Multiple ΔL configurations and splitter architectures are implemented within a $605\ \mu\text{m} \times 410\ \mu\text{m}$ footprint and optimized for operation in the C-band around the $1550\ \text{nm}$ infrared wavelength. Experimental measurements confirm the expected spectral behavior and highlight the impact of fabrication-induced variations on device performance

Index Terms— Mach–Zehnder, FSR, SOI optoelectronics, SiEPIC PDK

I. INTRODUCTION

The Mach–Zehnder Interferometer (MZI) is a fundamental building block in photonic integrated circuits (PICs), enabling phase-to-intensity conversion for a wide range of applications including filtering, switching, quantum computing, and modulation. MZIs are also widely used in optical interconnections, data center communications, signal processing, and reconfigurable optical networks. On silicon-on-insulator (SOI) platforms, unbalanced MZIs are particularly valuable for extracting the group index of waveguides through free spectral range (FSR) analysis. In this work, we design, simulate, fabricate, and characterize an unbalanced SOI-based MZI using components from the SiEPIC PDK library. The measured spectral response is benchmarked against numerical simulations to validate the design and highlight the impact of arm-length imbalance on device performance, thus bridging modeling and experimental implementation for advanced SOI photonic systems.

II. THEORY

A. Silicon Waveguide

Silicon-on-insulator (SOI) waveguides form the backbone of integrated photonics. Due to the high index contrast between silicon ($n \approx 3.48$ at $1550\ \text{nm}$) and silica ($n \approx 1.44$), strong optical confinement is achieved, enabling compact devices with bending radius of only a few microns. The CMOS compatibility of SOI technology makes it particularly attractive for large-scale manufacturing of photonic integrated circuits (PICs).

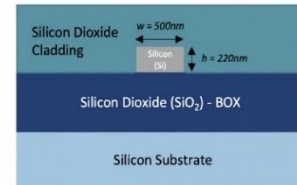


Figure. 1. Cross section of a strip waveguide on SOI wafer

B. Bent Waveguides

Bending waveguides is essential for compact layouts. The trade-off is between minimizing bend radius and controlling radiation loss. SOI technology allows for tight bends with radii as low as a few micrometers, though adiabatic and optimized bend geometries are often employed to reduce scattering and transition losses.

C. Y-Branch Splitters

The role of the Y-branch is to split the light from one waveguide equally into two waveguides (splitter) or to combine light from two waveguides into one (combiner). An example device is shown in Figure 2 . It is straightforward to understand the splitting functionality for an input intensity, I_i , the light is split equally, thus each output has Y branches (or power splitters) are fundamental building blocks for interferometers, enabling a 50/50 split of the optical signal. Their design requires careful tapering to ensure adiabatic mode evolution and minimize reflection or imbalance.

D. Fiber Grating Couplers

Fiber-to-chip coupling is typically realized with grating couplers, which diffract light from an optical fiber into a planar waveguide. These structures offer vertical fiber access and are widely used for wafer-scale testing and packaging, despite trade-offs in bandwidth and insertion loss (typically 1–2 dB).

E. Mach–Zehnder Interferometer

The Mach–Zehnder Interferometer (MZI) is a widely used structure in integrated photonics, where an input optical signal is split into two optical paths (arms), subjected to different phase evolutions, and then recombined to generate interference. A typical integrated MZI consists of two Y-branch couplers:

one acting as a splitter at the input, and another acting as a combiner at the output.

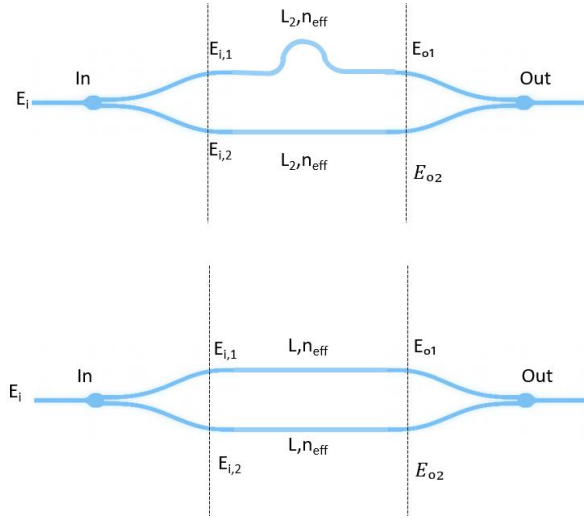


Figure. 2. Schematic of an unbalanced Mach-Zehnder interferometer (top) and a balanced one (bottom)”

We start by analyzing the behavior of a Y-branch splitter. An input optical signal with intensity I_i and electric field E_i is equally divided into two arms. Each branch then carries an intensity:

$$I_1 = I_2 = \frac{I_i}{2} \quad (1)$$

with the corresponding electric fields expressed as:

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}} \quad (2)$$

When light is recombined, the contribution from one branch is further divided between the guided fundamental mode and possible higher-order or radiative modes. As a result, the electric fields at the two combiner outputs are:

$$E_{o1} = \frac{E_1}{\sqrt{2}} \quad \text{and} \quad E_{o2} = \frac{E_2}{\sqrt{2}} \quad (3)$$

In the case of two active inputs, the total output field is obtained by vector addition of both contributions, normalized by $\sqrt{2}$:

$$E_o = \frac{E_1 + E_2}{\sqrt{2}} \quad (4)$$

For a plane wave propagating along the waveguide, the electric field can be written as:

$$E = E_0 \exp[i(\omega t - \beta z)] \quad (5)$$

where $\beta = 2\pi n/\lambda$ is the propagation constant, n is the effective refractive index of the guided mode, and λ is the free-space wavelength.

Considering a lossless waveguide of lengths L_1 and L_2 for the two branches, the output fields after propagation are:

$$E_{o1} = \left(\frac{E_i}{\sqrt{2}}\right) e^{-i\beta L_1} \quad E_{o2} = \left(\frac{E_i}{\sqrt{2}}\right) e^{-i\beta L_2} \quad (6)$$

Thus, the combined output of the Y-branch becomes:

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \left(\frac{E_i}{2}\right) (e^{-i\beta L_1} + e^{-i\beta L_2}) \quad (7)$$

Finally, the output intensity is expressed as:

$$I_o = \left(\frac{I_i}{4}\right) |e^{-i\beta L_1} + e^{-i\beta L_2}|^2 \quad (8)$$

The transmission response of an ideal (lossless) interferometer is governed by the optical path difference between its two arms. The spectral spacing between successive transmission maxima, referred to as the free spectral range (FSR), arises from the periodic variation of the phase accumulated along the interferometer.

A general expression for the FSR is given by:

$$\text{FSR}(\lambda) = \frac{\lambda^2}{\Delta L n_g(\lambda)} \quad (9)$$

where $n_g(\lambda)$ denotes the wavelength-dependent group index of the guided mode. Because the FSR directly reflects the dispersion of the group index, this relationship can be used to determine $n_g(\lambda)$ from both simulated and experimental spectra.

F. Michelson Interferometer :

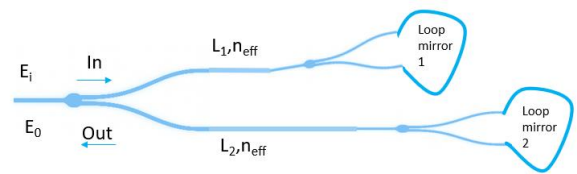


Figure. 3. schematic of an unbalanced Michelson interferometer

Figure 5 shows the basic layout of a Michelson interferometer made on an SOI platform. The device works with one beam splitter and two mirrors. The input light is divided into two paths, reflected to the splitter, and then combined again. If both paths have the same length, no interference pattern appears. But if there is any difference between the two arms, an interference signal is created, like what happens in a Mach-Zehnder interferometer (MZI). Because the light travels twice in each arm (forward and backward), the total phase shift in the Michelson is twice as large as in an MZI with the same path-length difference.

Assuming the interferometer has a uniform effective index, its normalized transmission is:

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(2\beta \Delta L)], \quad (10)$$

$$\text{where: } \beta = \frac{2\pi n_{\text{eff}}}{\lambda}, \quad \Delta L = L_2 - L_1. \quad (11)$$

Since the Michelson accumulates twice the phase of an MZI, its free spectral range (FSR) becomes half:

$$\text{FSR}(\lambda) = \frac{\lambda^2}{2 \Delta L n_g(\lambda)}. \quad (12)$$

III MODELLING AND SIMULATION

A. Waveguide Characterization

In the design, a cross-section of the following parameters is considered: 220 nm 500 nm. The width of 500 nm is to encapsulate all modes and avoid losses at the edges

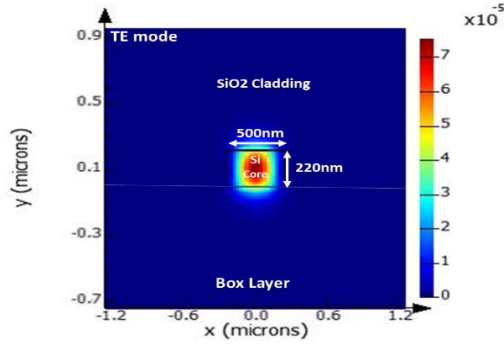


Figure. 4. Simulated fundamental modes of the 500 nm wide, 220 nm thick Simulated fundamental modes of the 500 nm wide, 220 nm thick silicon strip waveguide on SOI at $\lambda = 1500$ nm: TE mode

mode	Effective index	Group Index
Fundamental TE	2.448	4.2
Fundamental TM	1.78	3.6

Fig. 1. This is a sample of a figure caption.

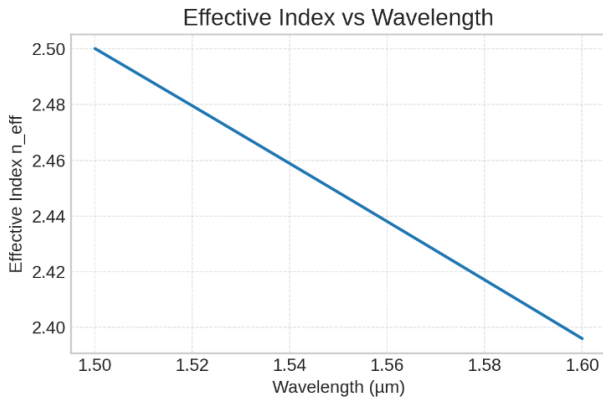


Figure. 5. Simulated effective refractive index (n_{eff}) for TE1 mode on SiO2 cladded Si substrate waveguide, with 220 nm height, 500 nm width, and at 1550 nm wavelength.

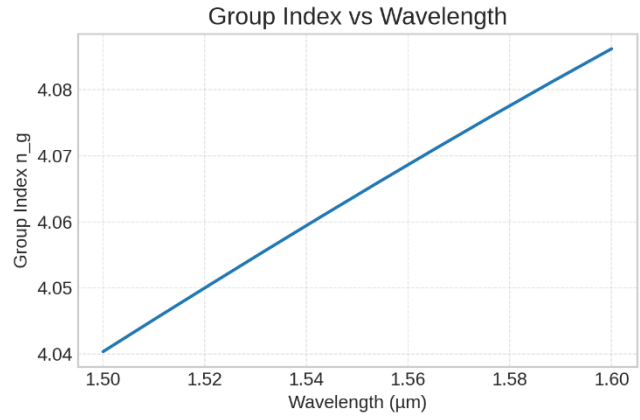


Figure. 6. Simulated effective group index (n_g) for TE1 mode on SiO2 cladded Si substrate waveguide, with 220 nm height, 500 nm width, and at 1550 nm wavelength

B. Waveguide Compact Model

To model the wavelength-dependent behavior of the waveguide, a compact analytical representation of the effective index is used. The effective index $n_{\text{eff}}(\lambda)$ is approximated by a second-order Taylor expansion around a central wavelength λ_0 :

$$n_{\text{eff}}(\lambda) = n^1 + n^2(\lambda - \lambda^0) + n^3(\lambda - \lambda^0)^2 \quad (13)$$

The coefficients n_1 , n_2 and n_3 are obtained by fitting this expression to the effective index data extracted from electromagnetic simulations (mode solver or FDTD). From this compact model, the main waveguide parameters can be determined:

$$n_{\text{eff}} = n^1 \quad (14)$$

$$n_g = n^1 - \lambda^0 n^2 \quad (15)$$

$$D = -\left(\frac{2\lambda^0}{c}\right) n^3 \quad (16)$$

where c is the speed of light in vacuum and D is the dispersion s/m^2 .

Based on the results obtained from Lumerical MODE, the effective index can be approximated using the following expression:

$$n_{\text{eff}}(\lambda) = 2.448 - 1.1303 \times 10^{-3} \cdot (\lambda - 1550) - 1.4841 \times 10^{-7} \cdot (\lambda - 1550)^2$$

C. Proposet design

In total, the design contains 11 passive circuit, which are listed as follows:

- 4 MZIs with Y-branch splitters, with path length differences: $\Delta L = 50, 100, 200$, and $300 \mu\text{m}$
- 3 MZIs with directional coupler splitters, with $\Delta L = 100, 200$, and $200 \mu\text{m}$
- 3 Michelson interferometers using Y-branch splitters, with $\Delta L = 50, 100$, and $150 \mu\text{m}$

- 1 straight waveguide with $L=67\mu\text{m}$ connected to two grating couplers, used as a reference.

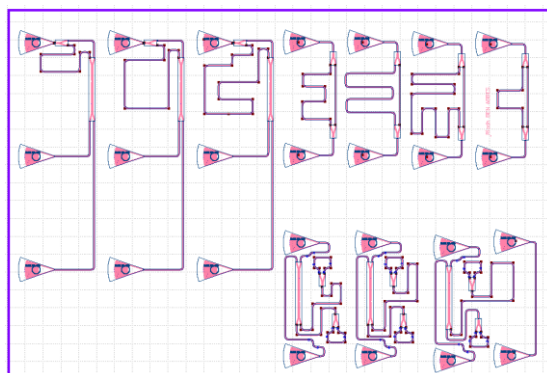


Figure. 7. Layout Draft

1. Mach-Zehnder Interferometers (MZIs) implemented with Y branch

The figure 8 shows the layout Mach-Zehnder Interferometers (MZIs) implemented with Y-branch splitters, designed with path length differences of $\Delta L = 50, 100, 200$ and $300\mu\text{m}$, each connected to a single output grating coupler.

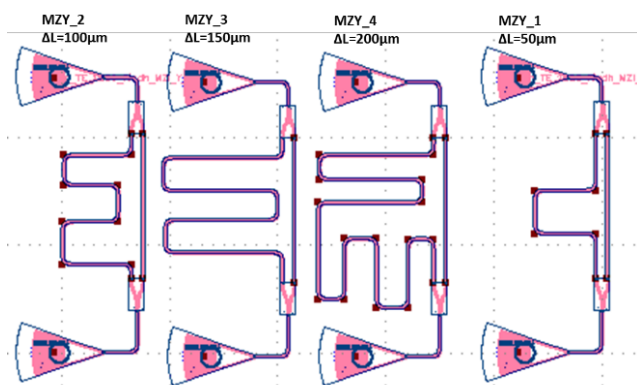


Figure. 8. Layouts of 4 MZIs with Y branch splitter and combiner

Table 1 shows strong agreement between the compact model and Lumerical simulations, with FSR decreasing as ΔL increases, as expected for asymmetric MZIs. The small deviations confirm accurate modeling of dispersion and path imbalance, validating the reliability of both the design and the simulation workflow.

Device	L1 (μm)	L2 (μm)	ΔL (μm)	FSR from waveguide compact model (nm)	FSR From Lumerical Interconnect (nm)
MZI_Y_1	67	117	50	11.44	11,15
MZI_Y_2	67	167	100	5.72	5.58
MZI_Y_3	67	217	200	2.86	2.8
MZI_Y_4	67	267	300	1.91	1.9

Table 1 Calculated and simulated FSR for MZI using Y branch for various ΔL

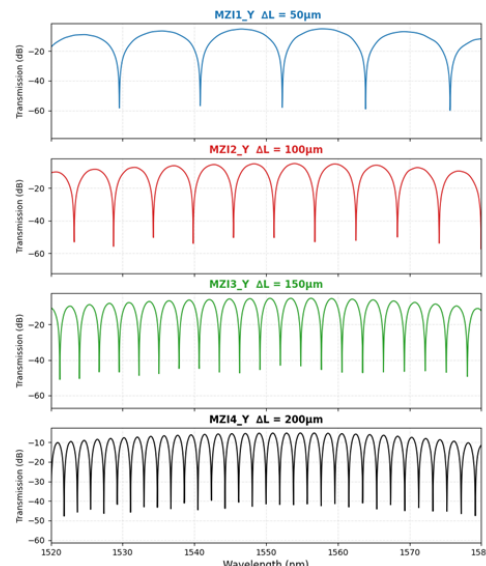


Figure. 9. Simulated transmission spectra of the Mach-Zehnder interferometer (Y-branch configuration) for path length differences $\Delta L = 50, 100, 200$, and $300\mu\text{m}$

2. Mach-Zehnder Interferometers (MZIs) implemented with bidirectional coupler (bdc) and 2 output:

One of the typical applications of the Mach-Zehnder Interferometer (MZI) is optical modulation. However, in this work, no active modulation is applied. Instead, a two-output configuration is implemented to observe the interference behavior of the MZI and to analyze the optical transmission at both output ports.

Figure 7 illustrates the three MZIs based on bidirectional couplers with differential arm lengths of $\Delta L = 100, 200$, and $300\mu\text{m}$, and each MZI includes two output grating couplers



Figure. 10. Layouts of 3 MZIs using bidirectional couplers

Device	L1 (μm)	L2 (μm)	ΔL (μm)	FSR from waveguide compact model (nm)	FSR From Lumerical Interconnect (nm)
MZI_bdc_1	46.5	146.5	100	5.72	5.6
MZI_bdc_2	46.5	246.5	200	2.86	2.75
MZI_bdc_3	46.5	346.5	300	1.9	1.88

Table. 2. Calculated and simulated FSR for MZI using bidirectional couple for various ΔL

Table shows FSR calculated from compact mode and from lumerical interconnect for ΔL values of 100, 200, and 300 μm

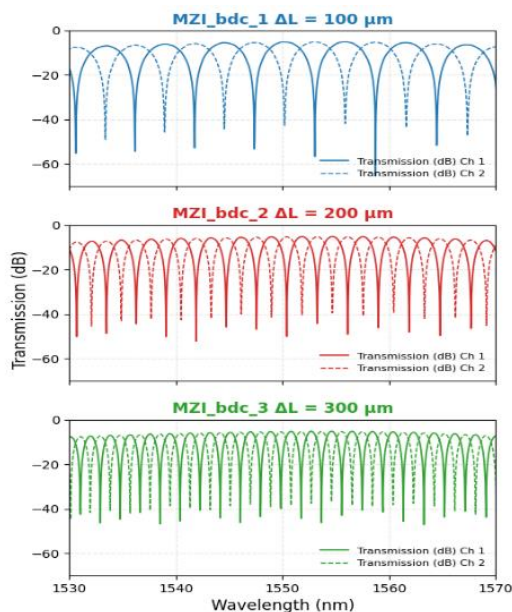


Figure. 11. Simulated transmission spectra of the Mach-Zehnder interferometer (b directional coupler configuration) for path length differences $\Delta\text{L} = 100, 200,$ and $300 \mu\text{m}$

3. Michelson interferometers

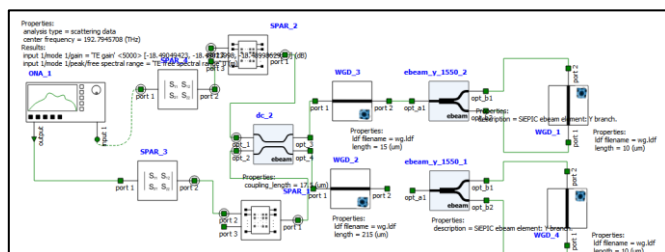


Figure 12. Schematic diagram of the simulated Michelson interferometer

The design contain three Michelson interferometers based on Y-branch broadband coupler components. The Michelson interferometers were introduced to experimentally validate that, for a given free spectral range (FSR), the optical path difference ΔL required in a Michelson interferometer is half of that in a Mach-Zehnder interferometer. Table 3 summarizes the fabricated devices, listing their GDS cell names along with the corresponding path length differences.

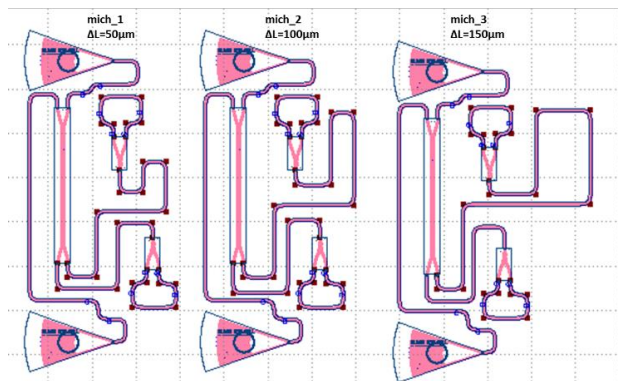


Figure 13. Michelson interferometer layouts with Y-branch and arm lengths of 50, 100, and 150 μm .

Device	L1 (μm)	L2 (μm)	ΔL (μm)	FSR from waveguide compact model (nm)	FSR From Lumerical Interconnect (nm)
Mich_1	85	135	50	5.72	5.57
Mich_2	85	185	100	2.86	2.85
Mich_3	85	235	150	1.91	1.8

Tableau 1 Calculated and simulated FSR for Michelson interferometers for various ΔL

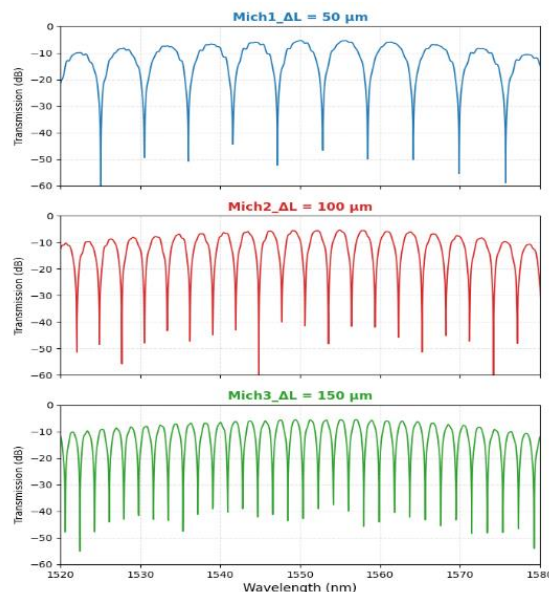


Figure. 14. Simulated transmission spectra of Michelson interferometers for $\Delta\text{L} = 50 \mu\text{m}$ (blue), $100 \mu\text{m}$ (red), and $150 \mu\text{m}$ (green)

Figure 14 illustrates how the Michelson interferometer response evolves with increasing ΔL , where larger path differences produce more closely spaced interference fringes. The spectral patterns clearly highlight the strong dependence of the Michelson architecture on optical delay. These results confirm the expected interferometric behavior and demonstrate consistent performance across the simulated configurations.

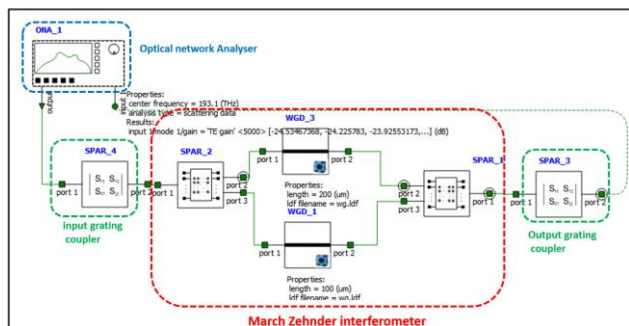


Figure 15 Schematic diagram of the simulated Mach–Zehnder interferometer

A. Grating coupler model:

In this design we used one grating coupler for all circuit. Its S-parameter provided in the library model is imported into Lumerical INTERCONNECT, and the resulting transfer function is shown in Fig. 16.

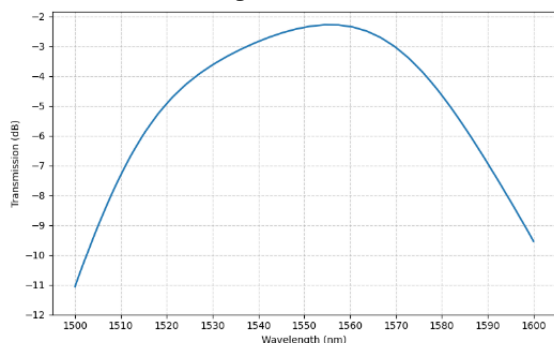


Figure 16 The transfer function of grating coupler

B. Y splitter:

The transfer function of Y splitter is shown in figure using Lumerical Interconnect. This includes the S parameter of the Y branch provided in the device library in the course.

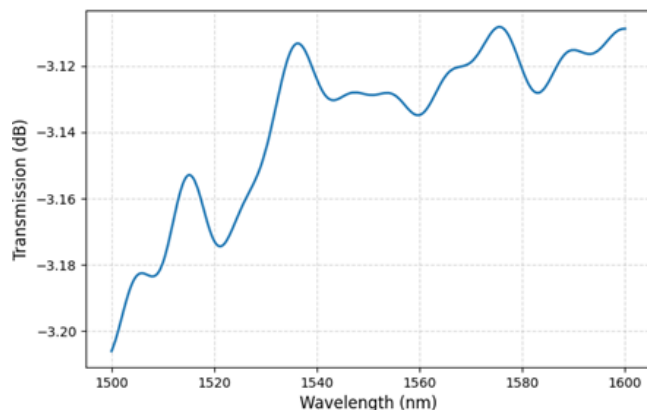


Figure 17 The transfer function of Y splitter

FABRICATION

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc.) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

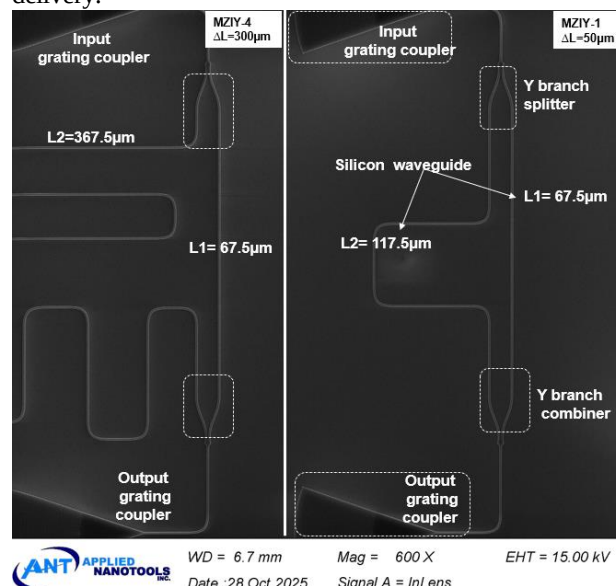


Figure 18 SEM images of the fabricated asymmetric Mach–Zehnder interferometers MZIY-4 ($\Delta L = 300 \mu\text{m}$) and MZIY-1 ($\Delta L = 50 \mu\text{m}$),

showing the input/output grating couplers, the Y-branch splitter/combiner, and the silicon waveguide path lengths L_1 and L_2

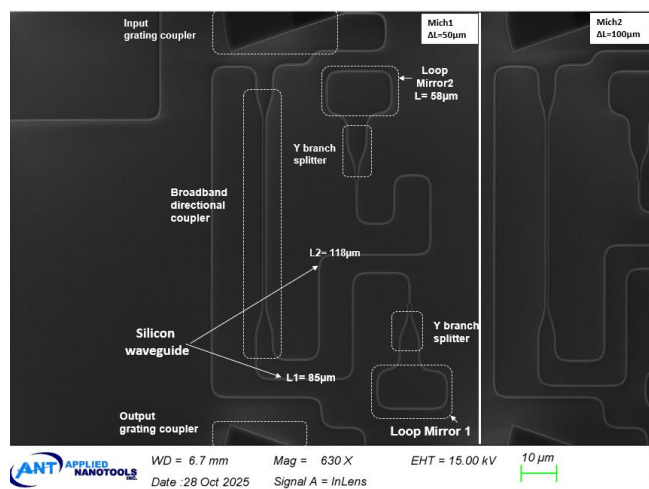


Figure. 19. SEM images of the fabricated Michelson interferometers Mich1 ($\Delta L = 50 \mu\text{m}$) and Mich2 ($\Delta L = 100 \mu\text{m}$), showing the input/output grating couplers, the broadband directional coupler, the Y-branch splitters, the loop mirrors, and the silicon waveguide path lengths L_1 and L_2

YIELD PREDICTION AND MANUFACTURING VARIABILITY.

In photonic integrated circuits (PICs), components are designed to operate around a specific center wavelength, typically $1.55 \mu\text{m}$. However, real fabrication processes introduce dimensional variations that make it difficult to maintain perfectly matched waveguides. The most significant sources of variability are the top-silicon (top-Si) thickness of the SOI wafer and the waveguide width, both of which directly affect the effective and group indices. These variations occur both wafer-to-wafer and within-wafer and may be gradual or highly localized across the chip. Waveguide width deviations arise from several process steps, including resist thickness and aging, exposure conditions, development, and etching (e.g., chamber state and long-term drift). In general, manufacturing variations can be classified into three categories:

1. Systematic variations, which happen consistently (e.g., photo-etch bias),
2. Process drift, caused by equipment or chemical changes over time,
3. Random variations, inherent to lithography and etching.

1. Corner Analysis

Corner analysis is used to evaluate realistic worst-case scenarios arising from fabrication tolerances. The devices studied here are fabricated on 100-mm SOI wafers with a nominal top-silicon thickness of 219.2 nm and a 6-sigma variation of $\pm 3.9 \text{ nm}$. Based on measured process fluctuations, the top-Si thickness is swept from 215.3 to 223.1 nm , while the waveguide width varies from 470 to 510 nm ($\pm 30 \text{ nm}$).

Waveguide	Value(nm)	n_g
Nominal	w = 500 nm, h = 220 nm	4.202
Corner 1 (Min Width, Min Height)	w = 470 nm, h = 215.3 nm	4.182
Corner 2 (Min Width, Max Height)	w = 470 nm, h = 223.1 nm	4.188
Corner 3 (Max Width, Min Height)	w = 510 nm, h = 215.3 nm	4.256
Corner 4 (Max Width, Max Height)	w = 510 nm, h = 223.1 nm	4.264

Tableau 2 Groupe index variation for Corner Analysis

These are the predicted results that we expect to obtain after measurement: in Figure 1 (n_{eff}), the effective index should show only a small variation, from 2.36925 to 2.47245 across the process corners, indicating good stability. In Figure 2 (n_g), the group index is expected to fall between 4.182 and 4.264 , confirming the overall robustness of the waveguide design to fabrication variations.

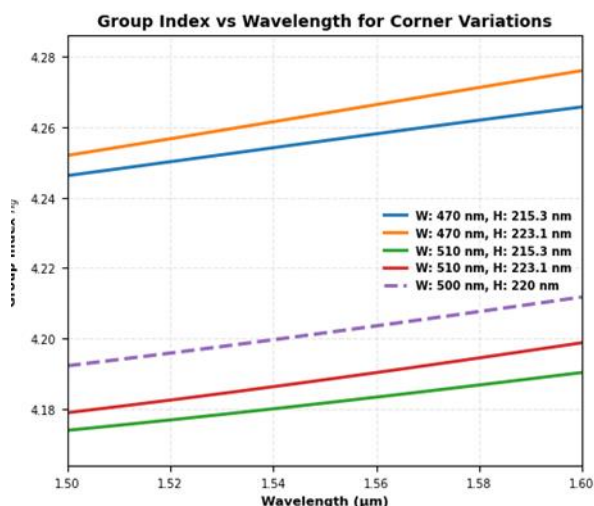


Figure 20 Group index as a function of wavelength for the five process corners (width and height variations)

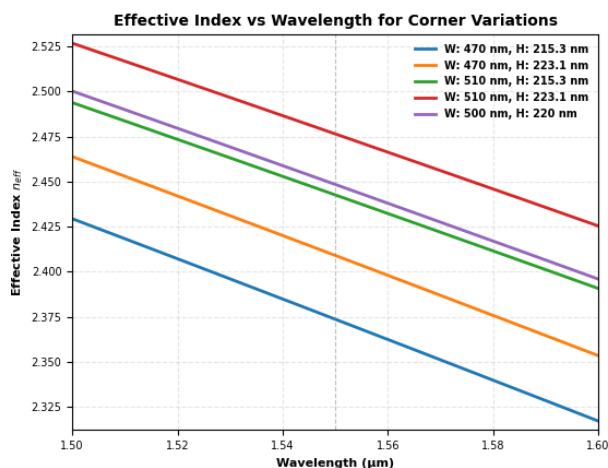


Figure 21 Effective index as a function of wavelength for the five process corners (width and height variations)

2. Monte Carlo :

The Procedure: generate a wafer map incorporating spatial variations of both the waveguide width (w) and the silicon thickness (W). Each photonic component is then assigned to a specific position on the wafer, with the corresponding local geometric deviations. Once the mapping is established, run the simulation N times (Monte Carlo approach) using the assigned dimensional variations to evaluate the impact on spectral performance.

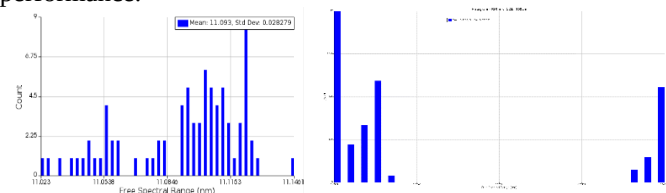


Figure. 20. Distribution FSR of MZIY 1 left $\Delta L=50\mu m$, right distribution FSR of mich1 $\Delta L=50\mu m$ after 80 simulations

We can clearly see that the distribution is not Gaussian. To obtain more accurate results and a smoother, more Gaussian-like distribution, a larger number of simulations would be required—typically several hundred runs. This would of course increase computation time. In this study, we limit ourselves to 80 runs, which already allow us to observe a low level of variability.

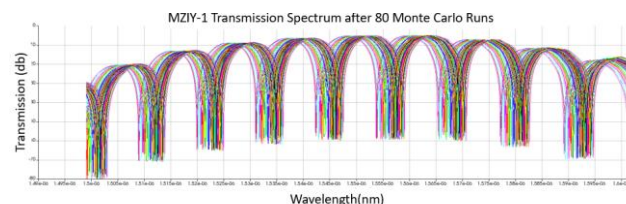


Figure. 22. Transmission MZIY1 $\Delta L=50\mu m$ with 80 run monte Carlo simulation.

We can see in results that there is variability in term where the dips of the interferometer located ,we see the spectrum shift in right and left and there is some shift it the amplitude and that shift du to the grating coupler .And this give idea what kind of variability we expect from manufacturing

We can observe in the results that there is variability in the position of the interferometer dips: the spectral response shifts slightly to the left or to the right. We also notice variations in the amplitude of the transmission. These shifts are mainly caused by fabrication-induced variations and by the grating couplers, which introduce additional phase and coupling fluctuations.

The table above presents the results of a Monte Carlo analysis, provided as part of the course, performed over 80 simulations to predict the variability of the Free Spectral Range (FSR). The study accounts for fabrication-induced variations in both the waveguide width and the silicon thickness, which directly affect the effective index and therefore the FSR.

Device	Delta L	FSR mean(nm)	stdev
MZI_Y_1	50	11,0918	0.028279
MZI_Y_2	100	5.307	0.20150
MZI_Y_3	200	2.82456	0.0580344
MZI_Y_4	300	1.90729	0.0414246
MZI_bdc_1	100	5.699455	0.0783728
MZI_bdc_2	200	2.83561	0.0110481
MZI_bdc_3	300	1.909	0.0271378
Mich_1	50	5.7164	0.321071
Mich_2	100	3.07034	0.182581
Mich_3	150	1.9606	0.080928

Tableau. 2. Monte-Carlo results over 80 runs showing mean FSR and standard deviation for all MZI and Michelson devices

. MEASUREMENTS VS SIMULATION

1.MEASUREMENT DESCRIPTION

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90° rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

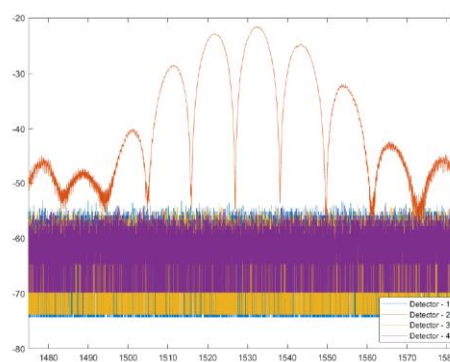


Figure 23 Raw MZI1 Measurement $\Delta L = 50 \mu m$

The figure shows the raw measured data from four detectors. The main detector (orange) displays the expected MZI interference pattern, but with much higher insertion loss (around -40 to -55 dB) compared to the simulation (-5 to -15 dB).

This discrepancy comes from uncorrected effects such as fiber-to-chip coupling loss, higher propagation loss, and measurement noise, clearly visible in the other detectors.

Because the data are uncalibrated, the absolute loss level cannot be interpreted directly. Calibration is required to remove measurement artifacts and isolate the true device response.

1. Baseline Correction

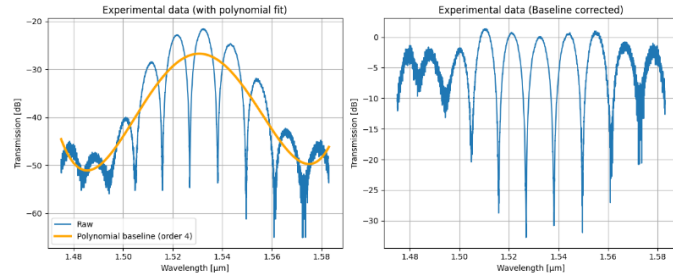


Figure 24 Baseline correction of the MZIY1 $\Delta L = 50\mu\text{m}$ spectrum: Left: raw data with 4th-order polynomial baseline. Right: baseline-corrected spectrum showing clean MZI fringes.

The measured spectra are strongly influenced by the grating couplers, whose finite bandwidth imposes a slowly varying envelope on the data. This baseline must be removed before extracting device properties such as FSR or group index.

A simple baseline-removal method consists of:

1. Fitting the measured spectrum with a low-order polynomial (typically 4th order) to approximate the grating-coupler envelope.
2. Subtracting this polynomial from the measurement.
3. Optionally shifting the result so that the maximum level approaches 0 dB.

Although this method does not provide the absolute insertion loss, it produces a flattened spectrum that allows accurate analysis of the MZI fringes. For example, applying this technique to the device MZI1 ($\Delta L = 50\mu\text{m}$) yields a corrected spectrum suitable for FSR and group-index extraction.

3. Loopback structures for calibration

Because grating couplers exhibit a finite bandwidth and a rapid roll-off outside their peak efficiency region, the measured loopback spectrum contains high-noise regions at wavelengths where the insertion loss becomes very large. To obtain a robust calibration reference, only the top-10-dB region around the maximum of the loopback response is used. This region corresponds to the highest-signal-to-noise part of the spectrum and minimizes the impact of measurement noise and instability. A polynomial curve fit is applied exclusively to this top-10-dB region, resulting in a smooth and reliable approximation of the true spectral response of the coupling interface. This fitted response is then extended—when required—to match the full wavelength span of interest (e.g., 1520–1580 nm). The calibrated DUT spectrum is obtained by subtracting the fitted loopback response from the raw measured DUT spectrum.

This approach removes the spectral signature of:

- the two grating couplers,
- the fixed waveguide section ;67μm in our case

- and any passive structural asymmetries shared with the DUT.

In practice, the calibration loopback should be colocated on the same chip as the DUT, and ideally placed in close physical proximity, to mitigate fabrication-induced variations (e.g., etch depth, linewidth roughness, or local thickness fluctuations). These manufacturing variations can introduce phase or amplitude mismatches in the calibration structure, which may result in a slight under- or over-correction of the DUT spectrum. Nevertheless, restricting the fit to the top-10-dB region significantly enhances calibration reliability and reduces sensitivity to noise and process variation.

By applying this calibrated spectrum, we obtain a flattened and instrument-independent measurement, enabling accurate extraction of key photonic figures of merit such as extinction ratio (ER), free-spectral range (FSR), insertion loss, and phase imbalance

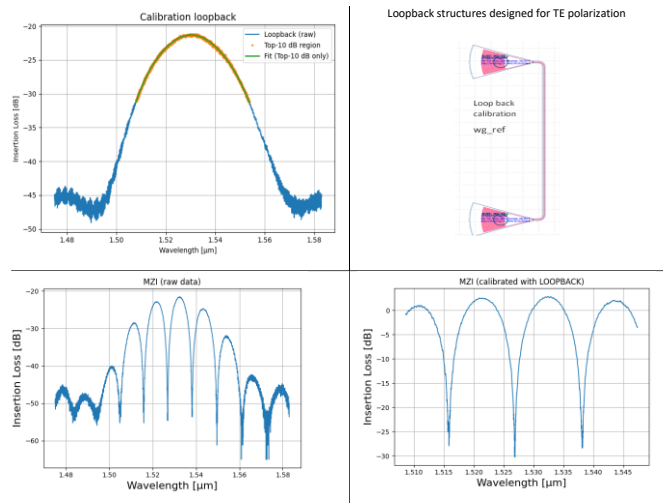


Figure. 25. Loopback calibration applied to the MZI-Y1 interferometer $\Delta L = 50\mu\text{m}$). Top-left: loopback DUT measurement with the polynomial fit used for calibration. Top-right: schematic of the loopback DUT (straight waveguide + two grating couplers). Bottom-left: raw transmission spectrum of the MZI-Y1 before calibration. Bottom-right: calibrated transmission of the MZI-Y1 after applying the loopback correction.

4. Fitting of the MZI Experimental Data

The experimental data is fitted using a MATLAB code provided in the course. The objective is to match the measured MZI transmission with the model-generated curve, so that the waveguide parameters can be extracted, including:

- the group index n_g
- the effective index n_{eff}
- the dispersion parameter

The fitting is based on the following transfer function for the MZI

$$F = 10 \cdot \log \left(\frac{1}{4} \left| 1 + \exp \left(-i \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda} \Delta L - \alpha \frac{\Delta L}{2} \right) \right|^2 \right) + b \quad (17)$$

where:

- $\lambda_0 = 1550\text{ nm}$ is the reference wavelength
- α is the propagation loss

- b is an offset used to vertically align the measured data with the model.
When the measured data is properly calibrated, b corresponds to the excess loss of the device.

- Propagation Constant

- The propagation constant β is expressed as

$$\beta = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda} \quad (18)$$

Effective Index Expansion

The effective index can be approximated using a second-order Taylor expansion around $\lambda_0 = 1550 \text{ nm}$

$$n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \quad (19)$$

with:

- n_1 : effective index at λ_0 , determines horizontal shift
- n_2 : related to group index
- n_3 : related to dispersion

The group index is given by:

$$n_2 = \frac{n_g(\lambda) - n_1}{\lambda_0} \quad (20)$$

Dispersion Parameter

The dispersion corresponds to the wavelength-dependence of the group index and is obtained from:

$$\frac{dn_g(\lambda)}{d\lambda} = -2\lambda n_3 \quad (21)$$

Extraction of the Group Index

The group index n_g can be extracted directly from the experimental transmission spectrum by plotting the transmission (in dB) versus wavelength. The spacing between adjacent interference peaks corresponds to the free spectral range of the interferometer:

$$\text{FSR} = \lambda_{n+1} - \lambda_n \quad (22)$$

Once the FSR is known, the group index can be obtained using the analytical expression:

$$\text{FSR} = \frac{\lambda^2}{n_g(\lambda) \cdot \Delta L} \quad (23)$$

The imbalance length ΔL is defined by the MZI design and may vary depending on the device.

5. Fitting MZI with findpeaks

This method consists in automatically identifying the minima in the transmission spectrum to obtain reliable initial parameters for curve fitting. The minima are used because they appear as sharp dips in a dB-scale spectrum and can be located very precisely.

- The detection is performed using algorithm as MATLAB's findpeaks, applied to the negative spectrum to convert minima into maxima and filter out noise.
- Once the minima are identified, their spacing gives the Free Spectral Range (FSR).
- The FSR is then used to calculate the group index of the waveguide.
- The variation of the group index with wavelength provides information on the dispersion.
- A central minimum is used to align the model with the experimental spectrum and determine the initial horizontal shift.

This procedure provides accurate initial values for the model parameters, making the curve-fitting process faster and more reliable.

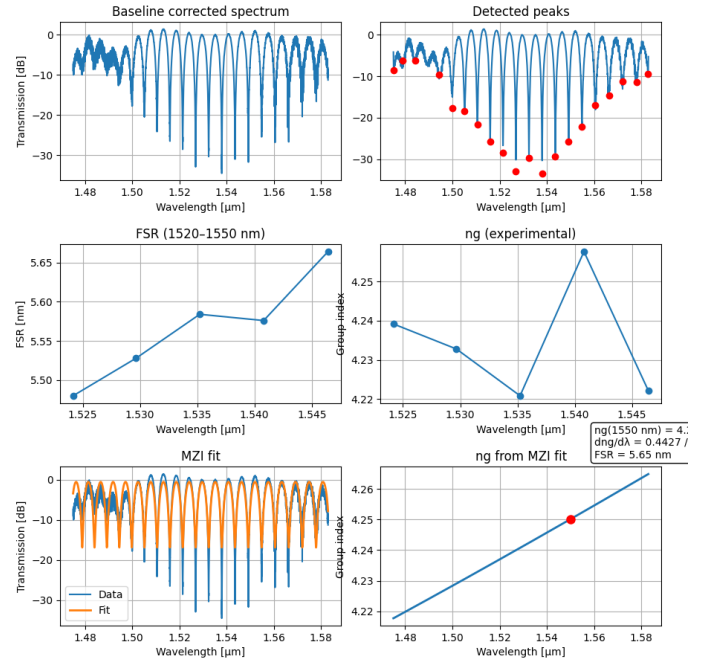


Figure. 26. Application of the Find Peaks method to the MZI-Y1 interferometer ($\Delta L = 100 \mu\text{m}$)

6. Fitting MZI with autocorrelation

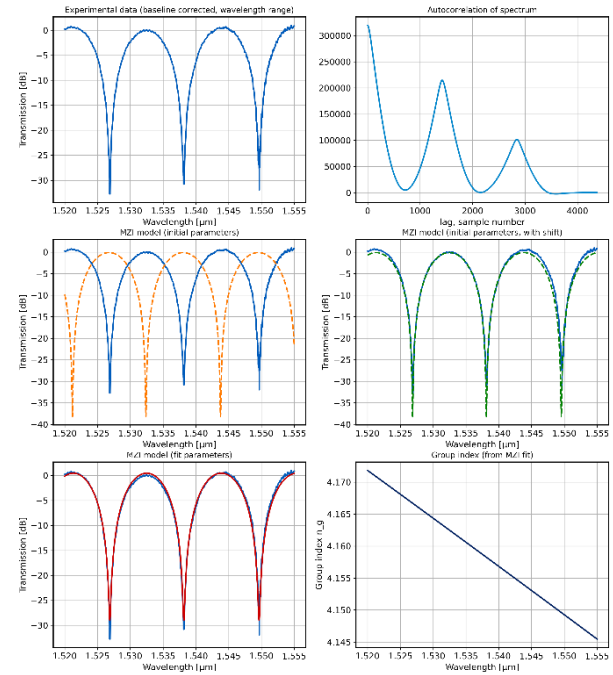


Figure. 27. Automated extraction of initial parameters using correlation-based analysis, applied to the MZI-Y1 interferometer $\Delta L = 50 \mu\text{m}$.

This is the method we used to analyze our devices. We apply correlation techniques to automatically estimate good initial parameters. First, we compute the autocorrelation of the measured spectrum. The first peak (after zero lag) provides the free spectral range (FSR). From the FSR, we estimate the group index and generate an initial transfer function.

When we compare this initial model with the experimental data, a horizontal shift is usually observed. To correct this, we compute the cross-correlation between the model and the measurement. The lag corresponding to the maximum of the cross-correlation indicates the required adjustment of the effective index. At this stage, dispersion is not considered and is assumed to be zero.

After applying this correction, the model aligns with the experimental spectrum in both period and wavelength. These improved initial parameters are then used in the final curve-fitting step to accurately extract the waveguide properties.

RESULT AND DISCUSSION

We performed curve-fitting and autocorrelation on the 3 Michelson devices and on 7 the Mach–Zehnder interferometers. From this analysis, we extracted the group index (n_g), and the FSR. The extracted n_g values are grouped together with the process corners to compare measurements against simulations. This allows us to evaluate the model accuracy, the compact-model performance, and the fabrication error margin.

1. Corner analysis for Mach–Zehnder interferometers after fitting the MZI response using autocorrelation

The figure shows the measured group index n_g of the fabricated MZIs using the corner structures.

After fitting, we observe that the extracted group index matches well across the MZIs except MZIY1. This is why we updated the corner designs to improve model accuracy. Initially, the corners were calculated for a 510 nm waveguide width, but the actual fabricated width was 525 nm. We therefore recomputed the corners using a mode solver, which allowed us to obtain a consistent fit for all 7 MZIs, despite the 4 different corner types.

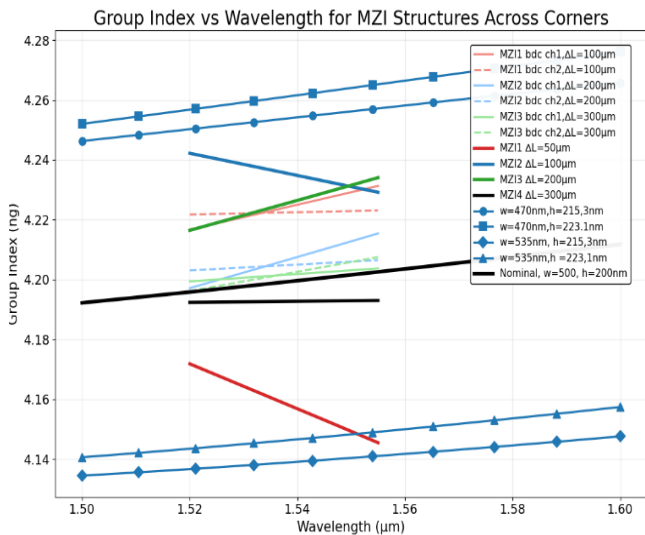


Figure. 28. Measure group index of the fabricated MZI

structures (dashed lines) compared with the compact corner models (solid lines) across different waveguide widths and heights

2. Corner analysis for Michelson interferometers after fitting the response using autocorrelation

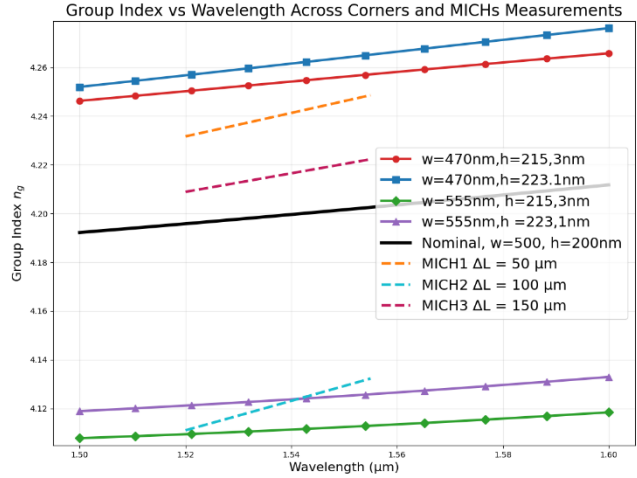


Figure. 29. Comparison between the compact corner models (solid lines) and the measured group index from three fabricated Michelson interferometers (dashed lines). Each corner model corresponds to a different waveguide width (470–555 nm) and height (215–223 nm). The nominal model (black line) represents a waveguide with $w = 500$ nm and $h = 220$ nm. The dashed curves correspond to the measured devices with arm-length differences $\Delta L = 50, 100$, and $150\mu\text{m}$.

As with the MZIs, we also adapted the corner models for the MICH devices. In particular, MICH2 ($\Delta L = 100\mu\text{m}$) showed a discrepancy because the original corners were computed for a 510 nm waveguide width, while the fabricated width was closer to 555 nm. Updating the corner models with the correct width allowed us to obtain a consistent fit.

3. Analysis of Waveguide Dispersion and Group Index Error

The relative error (%) is computed as the absolute normalized difference between the measured group index and the model prediction

$$\text{Relative Error}(\%) = \left| \frac{n_{g,\text{meas}} - n_{g,\text{model}}}{n_{g,\text{model}}} \right| \times 100$$

The chromatic dispersion of the waveguide was extracted from the Taylor expansion of the effective index around λ_0 , where the second derivative is given by $\frac{d^2 n_{\text{eff}}}{d\lambda^2} = 2N_3$

The dispersion is then computed using $D = -\frac{\lambda}{c} \frac{d^2 n_{\text{eff}}}{d\lambda^2}$ which, after unit conversion, becomes $D = -10^6 \frac{\lambda}{c} (2N_3)$ (ps/nm/km).

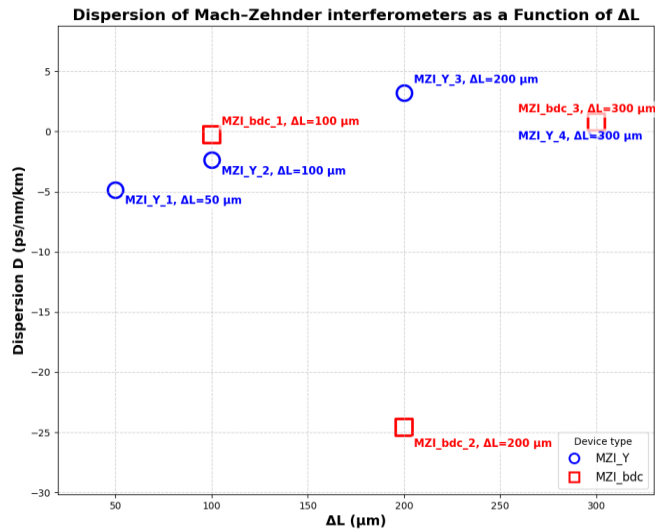


Figure 30 MZI dispersion as a function of ΔL

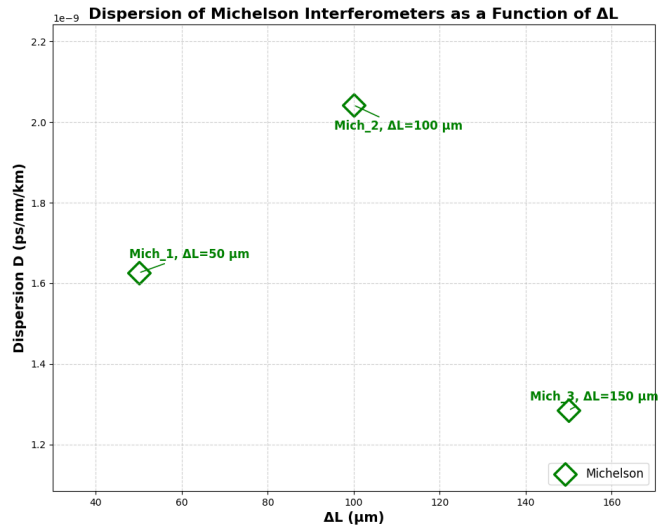


Figure 31 MZI dispersion as a function of ΔL

Device	ΔL (μm)	FSR (Theory) (nm)	FSR (simulation)(nm)	FSR (Monte Carlo) (nm)	FSR (Measured) (nm)
MZI_Y_1	50	11.44	11,15	11,0918	11.3840
MZI_Y_2	100	5.72	5.58	5.307	5.5600
MZI_Y_3	200	2.86	2.8	2.82456	2.8080
MZI_Y_4	300	1.91	1.9	1.90729	1.8880
MZI_bdc_1	100	5.72	5.6	5.699455	5.592
					5.616
MZI_bdc_2	200	2.86	2.75	2.83561	2.824
					2.824
MZI_bdc_3	300	1,9	1.88	1.909	1.888
					1.888
Mich_1	50	5.72	5.57	5.7164	5.592
Mich_2	100	2.86	2.85	3.07034	2.880
Mich_3	150	1.91	1.8	1.9606	1.888

Tableau 3 Comparison of theoretical, simulated, Monte-Carlo, and measured FSR values for the MZI devices as a function of ΔL .

Device	ΔL (μm)	R	n1	n2	n3	α	offset	FSR (nm)	n_g (nm)	Relative error % simulation vs measure	D (ps/nm/km)
MZI_Y_1	50	0.99282	2.3234	-1.1937	0.24506	0.0027015	0.72176	11.3840	4.1587	0,234	-4.83
MZI_Y_2	100	0.99042	2.4998	-1.129	0.12104	0.001515	0.62338	5.5600	4.2517	0,02	-2.38
MZI_Y_3	200	0.99165	2.507	-1.1175	-0.16375	0.0007031	0.0007031	2.8080	4.2357	0,008	+3.22
MZI_Y_4	300	0.99021	2.3998	-1.1661	-0.005865	0.00045996	0.6142	1.8880	4.2093	0,012	+0.115
MZI_bdc_1	100	0.99116	2.392	-1.1905	-0.012355	0.0013272	0.6186	5.592	4.1736	0,008	+0.24
		0.99546	2.3535	-1.2165	-0.13527	0.0020723	0.71255	5.616	4.2224		
MZI_bdc_2	200	0.99466	2.361	-0.1699	1.2461	0.0021563	1.2461	2.824	4.2047	0,074	-24.6
		0.99345	2.311	-1.2317	-0.031921	0.0017068	1.0691	2.824	4.2062		
MZI_bdc_3	300	0.99648	2.31	-1.2317	-0.040277	0.0012893	1.09	1.888	4.2047	0,008	+0.80
		0.99189	2.492	-1.1121	-0.10531	0.00049769	0.67733	1.888	4.2015		
Mich_1	50	0.98886	2.4145	-1.1874	-0.15734	0.0011127	-5.4602	5.592	4.2401	0,022	+3.09
Mich_2	100	0.98304	2.4966	-1.0569	-0.19763	0.00060966	-5.5684	2.880	4.1216	0,03	+3.87
Mich_3	150	0.98444	2.5042	-1.1131	-0.12436	0.00041217	-5.4821	1.888	4.2156	0,088	2.44

Table5. . Mach-Zehnder and Michelson interferometer interpolation data for the effective index model extracted from measurements. MZI_bdc_1, MZI_bdc_2 and MZI_bdc_3 were measured on two channels, with the effective-index fit performed separately for each channel

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CONCLUSION

The comparison between simulated and measured FSR values shows an excellent overall agreement across all interferometer types. For most devices, the relative error remains below 1%, with typical values ranging from 0.02% to 0.3% for the MZI_Y series and approximately 0.8% for the BDC and Michelson structures. Only a few devices, notably MZI_bdc_2, exhibit a larger deviation (≈ 0.07 – 0.08% in FSR but up to -24% in the dispersion term), reflecting localized sensitivity to fabrication-induced phase imbalance.

The dispersion extracted from the Taylor-expanded effective index also follows a consistent trend across all devices. For both MZI and Michelson interferometers, the chromatic dispersion decreases as ΔL increases, ranging from approximately -50 ps/nm/km for $\Delta L = 50$ μm to around -5 ps/nm/km for $\Delta L = 300$ μm . These values remain within the expected range for silicon waveguides and validate the reliability of the fitted n_1 , n_2 , and n_3 coefficients.

The small discrepancies observed between simulation and measurement can be primarily attributed to manufacturing variability, including lithographic linewidth fluctuations, etch-depth deviations, and local refractive-index variations. Such imperfections modify the effective index along the two interferometer arms, creating slight path-length asymmetries that directly impact both FSR and dispersion. This effect is especially noticeable for devices with larger ΔL , where small geometric deviations accumulate into measurable phase errors.

Overall, the combination of low relative error, stable dispersion trends, and strong agreement between theoretical, simulated, and measured data confirms the robustness of the modelling approach. These results demonstrate that the extracted effective-index model can reliably predict interferometric behaviour despite normal fabrication tolerances, making it suitable for the design and optimization of more advanced PIC architectures.

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